IGCSE 0607 Practice Questions — E1.3 — Combined Questions & Markscheme

Powers and roots

Total Questions: 3 — **Total Marks:** 16 Marks

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  → **Paper 2 (Non-Calculator)** *(2 Qs)* · **Paper 4 (Calculator)** *(1 Q)*

1 **(a)** Work out the value of

$$\sqrt{169} + 2^3$$

..... **[2]**

(b) Work out the value of

$$4^2 \times \sqrt[3]{125}$$

..... **[2]**

2 Solve the following equations.

(a)

$\sqrt[3]{3x+4} = 4$

$x =$ ..... **[3]**

(b)

$x^2 = 4^3 + 6^2$

$x =$ ..... , or $x =$ ..... **[2]**

(c)

$\sqrt{x} + 2\sqrt{x} = 21$

$x =$ ..... **[2]**

3 **(a)** Calculate the value of

$$\frac{\sqrt[3]{450} + 12.5^2}{3.4^3 - \sqrt{80}}$$

Write down all the figures on your calculator display.

..... **[2]**

(b) Write your answer to part (a) correct to 3 decimal places.

..... **[1]**

(c) Find the largest integer n such that $n^4 < 2000$.

$n =$ ..... **[2]**

1 **(a)** 21 **[B2]**

M1 for 13 or 8

(b) 80 **[B2]**

M1 for 16 or 5

[Back to Question] *[Total Marks: 4]*

2 **(a)** $3x + 4 = 64$ **[M1]**

$3x = 60$ **[M1]**

20 **[A1]**

(b) $x^2 = 64 + 36 = 100$ **[M1]**

$10, -10$ **[A1]**

(c) $3\sqrt{x} = 21$ or $\sqrt{x} = 7$ **[M1]**

49 **[A1]**

[Back to Question] *[Total Marks: 7]*

3 **(a)** 5.3990299... **[B2]**

M1 for numerator evaluated to 163.91... or denominator evaluated to 30.359...

(b) 5.399 **[B1]**

(c) 6 **[B2]**

M1 for evaluating $6^4 = 1296$ and $7^4 = 2401$

[Back to Question] *[Total Marks: 5]*